

Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

Meeting 2 (Extended): Reframing Faithfulness, Hallucination
Detection & Real-Time Metrics

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Date: 18th of August 2026
Version: 1.0

Agenda

- System Overview
- Performance Metrics
- Hardware & Model Detection
- Hallucination Detection
- Hallucination Scoring & Faithfulness
- Human Labeling Validation



System Overview

Performance Monitoring

- Measures time to first token (TTFT)
- Tracks tokens per second generation rate
- Calculates latency percentiles (P50, P95)
- Identifies performance bottlenecks

Quality Assessment

- Detects hallucinations and contradictions
- Scores response faithfulness to source
- Evaluates factual consistency

Model Analysis

- Detects hardware utilization (CPU/GPU)
- Identifies model quantization levels
- Calculates model size and resource needs



4 Performance Metrics

Performance Indicators

Metric	What It Measures	Why It Matters
TTFT	Time to first token	User perception of speed
Tokens/sec	Generation throughput	System capacity
P50	Median latency	Typical user experience
P95	95th percentile	Worst-case performance

Theoretical Interpretation

- **TTFT**: Lower is better
- **Tokens/sec**: Higher is better
- **P95**: Lower is better (reliability under load)
- **Pattern**: Better hardware = faster generation

What Affects Performance:

- Model size (larger = slower)
- Quantization level (lower bits = faster)
- Hardware (GPU vs CPU)

Hardware & Model Detection

Hardware Detection:

- **GPU Detection:**

- Identifies GPU model
- Measures available GPU memory
- Confirms CUDA availability

- **CPU Detection:**

- Detects CPU model
- Counts available cores
- Used when GPU is unavailable

Model Properties

Property	What It Tells Us
Quantization	Speed vs quality trade-off
Parameter Size	Model complexity
File Size	Storage/loading time
Context Length	Maximum input size

Hallucination Detection

Natural Language Inference (NLI):

NLI determines logical relationship between source text and generated text

Label	Definition	Interpretation
Entailment	Generated text follows logically	Faithful response
Contradiction	Directly conflicts with source	Hallucination
Neutral	Unclear or unrelated	Potential issue

Severity Calculation

$$\text{Severity} = (\text{Contradictions} \times 1.0) + (\text{Neutral} \times 0.5)$$

- Contradictions weighted (100%)
- Neutral weighted (50%)
- Threshold: > 0.3 triggers hallucination flag

Hallucination Scoring & Faithfulness

Faithfulness:

- How well response reflects source content
- Penalizes unsupported statements
- Rewards accurate restatement

Score	Meaning	Response Quality
Faithfulness Score	Alignment with source	Higher = better
Hallucination Severity	Degree of contradiction	Lower = better
Confidence	NLI model certainty	Higher = better

Faithfulness Score

Faithfulness = Entailment - (Contradictions × 0.5)

- Range: -0.5 to 1.0
- Positive score = mostly faithful
- Negative score = mostly hallucinated

Examples:

- Score 0.8 = Strongly faithful to source
- Score 0.3 = Some unsupported claims
- Score -0.2 = Significant contradictions

Human Labeling Validation

Human Labeling:

- Confirm statements provide truth
- Uses NLI scores to classify
- Labels: "faithful" or "hallucinated"
- Sets benchmark for AI performance

AI Prediction:

- Same categories as human

Metric	Definition	Interpretation
Accuracy	% of correct predictions	Overall performance
F1 Score	Balanced precision/recall	Handles imbalanced data
Confidence	AI's certainty	Reliability indicator

Metrics:

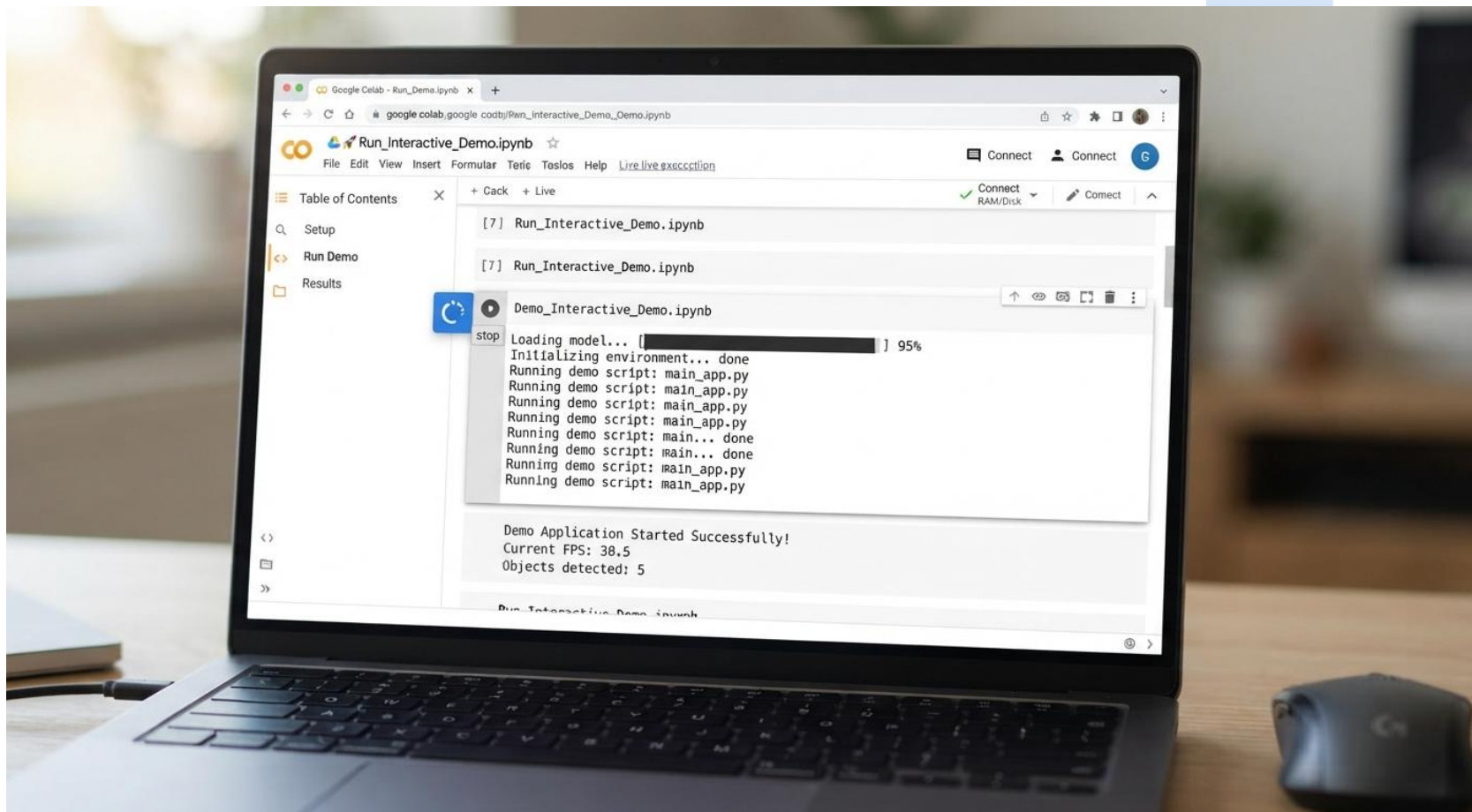
- **Accuracy:** Good when classes balanced
- **F1 Score:** Better when statement is hallucinations
- **Confidence:** Shows if AI is certain or uncertain

Validation Goal:

- Measure AI's alignment with human judgment
- Identify systematic biases

9

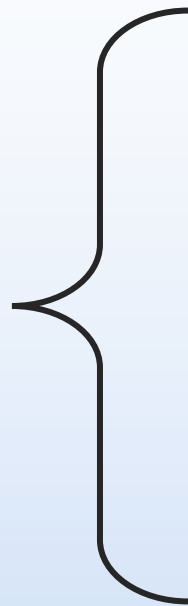
LIVE DEMO



References

- Time to First Token (TTFT) – <https://www.emergentmind.com/topics/time-to-first-token-ttft>
- P50/P95 – <https://qaskills.sh/blog/performance-test-percentiles-p95-p99-guide>
- Large Language Model (LLM) Quantization – <https://localllm.in/blog/quantization-explained>
- Accuracy & F1 Score – <https://www.datacamp.com/tutorial/f1-score>
- Scikit-learn “accuracy_score” – https://scikit-learn.org/stable/modules/generated/sklearn.metrics.accuracy_score.html
- Scikit-learn “f1_score” – https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html

Thank you



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